Module 4

Bayesian Computation and MCMC Methods

Department of Mathematics and Statistics
The Open University of Kenya

Module Overview

In Bayesian inference, the posterior distribution provides a complete description of uncertainty about model parameters. However, for complex or high-dimensional models, computing the marginal likelihood (evidence) to normalize the posterior is analytically intractable. This module introduces the computational foundations of modern Bayesian statistics: why analytical integration fails, how **Markov chains** construct stochastic processes that converge to a target distribution, and how **Markov Chain Monte Carlo (MCMC)** algorithms—specifically **Metropolis–Hastings** and **Gibbs sampling**—generate samples from complex posterior distributions without calculating normalizing constants. Finally, you will explore **MCMC convergence diagnostics** (trace plots, autocorrelation, Gelman–Rubin $\hat{R}$) and apply these techniques to real-world Kenyan case studies in public health, urban traffic flow, and credit risk assessment.

Expected Learning Outcomes

By the end of this module, you should be able to:

1. **Explain** the mathematical challenges of computing marginal likelihoods in high-dimensional Bayesian models.

2. **Define** Markov chains and verify key properties such as stationarity, ergodicity, and convergence.

3. **Derive** and apply the Metropolis–Hastings sampling algorithm and its symmetric proposal special case.

4. **Implement** Gibbs sampling for multivariate posterior distributions using full conditional distributions.

5. **Evaluate** MCMC convergence using trace plots, autocorrelation functions, and the Gelman–Rubin statistic ($\hat{R}$).

6. **Apply** MCMC methods and convergence diagnostics to practical problems using R code in real-world contexts.

1 Bayesian Computation Challenges

In Bayesian inference, the posterior distribution provides a complete description of uncertainty about the parameter θ given data y. However, computing this posterior is often mathematically challenging, especially in complex models.

Definition

The posterior distribution is given by:

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where the **marginal likelihood** (or **evidence**) is the normalizing constant:

$$p(y) = \int p(y \mid \theta)p(\theta)\, d\theta$$

- $p(y \mid \theta)$: Likelihood (data information)
- $p(\theta)$: Prior (initial belief)
- $p(y)$: Marginal likelihood (normalizing constant)

Key Idea

The marginal likelihood $p(y)$ ensures that the posterior distribution integrates to 1, but it is often extremely difficult or impossible to compute analytically.

1.1 Why Computation Becomes Difficult

In many practical problems, the integral defining $p(y)$ has no closed-form solution. This is especially true when:

- The model involves multiple parameters (high-dimensional parameter space)
- The likelihood function is complex or non-linear
- The prior distribution is non-conjugate

Key Idea

As model complexity increases, analytical solutions become infeasible, requiring numerical integration or simulation-based sampling methods.

1.2 High-Dimensional Integration Problem

Example 1. ***High-Dimensional Parameter Space***

Consider a Bayesian regression model with d parameters $\theta = (\theta_1, \ldots, \theta_d) \in \mathbb{R}^d$. The marginal likelihood becomes a d-dimensional integral:

$$p(y) = \int_{\mathbb{R}^d} p(y \mid \theta) p(\theta) \, d\theta$$

Interpretation: As the dimension d increases, the volume of the parameter space grows exponentially. Numerical quadrature rules (like Simpson's rule or grid evaluation) quickly become computationally impossible.

Key Idea

This exponential growth in computational complexity with the number of parameters is known as the **curse of dimensionality**.

1.3 Illustrative Example: Non-Conjugate Logistic Regression

Example 2. ***Non-Conjugate Logistic Model***

Suppose we model binary outcomes $y \in \{0, 1\}$ using logistic regression:

$$P(y = 1 \mid \theta) = \frac{1}{1 + e^{-\theta x}}$$

with a standard normal prior:

$$\theta \sim \mathcal{N}(0, 1)$$

Challenge: *The unnormalized posterior is:*

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta) = \left(\frac{1}{1 + e^{-\theta x}} \right)^y \left(1 - \frac{1}{1 + e^{-\theta x}} \right)^{1-y} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\theta^2}$$

This expression does not match any standard distribution because the Gaussian prior is not conjugate to the Bernoulli/logistic likelihood. Consequently, the marginal likelihood:

$$p(y) = \int_{-\infty}^{\infty} p(y \mid \theta) p(\theta) \, d\theta$$

cannot be evaluated analytically.

1.4 Practical Consequences

- Exact posterior computation is often impossible for modern statistical models.
- Statisticians rely on approximation methods:
  - Monte Carlo methods
  - Markov Chain Monte Carlo (MCMC)
  - Variational Inference

Key Idea

Modern Bayesian analysis depends heavily on computational algorithms and simulation rather than closed-form mathematical derivations.

2 Markov Chains

Markov chains form the foundation of MCMC methods. They provide a stochastic mechanism for generating samples from complex target distributions by constructing a sequence of random variables that evolves over time.

Definition

A **Markov chain** is a stochastic process $\{X_t\}_{t \geq 0}$ satisfying the **Markov property**:

$$P(X_{t+1} \mid X_t, X_{t-1}, \ldots, X_0) = P(X_{t+1} \mid X_t)$$

Interpretation: Given the present state X_t, the future state X_{t+1} is conditionally independent of past history $(X_{t-1}, \ldots, X_0)$. This memoryless property greatly simplifies modeling and computation.

Key Idea

Markov chains allow us to explore complex, high-dimensional probability distributions by making local, probabilistic transitions through the state space.

2.1 Transition Mechanism

A time-homogeneous Markov chain evolves according to a **transition probability**:

$$P(x, x') = P(X_{t+1} = x' \mid X_t = x)$$

This defines the probability of moving from state x to state x' in one discrete step.

Example 3. ***Two-State Weather Model***

Suppose weather is modeled as either Sunny (S) or Rainy (R). The transition probabilities are:

$$P(S \mid S) = 0.8, \quad P(R \mid S) = 0.2$$

$$P(S \mid R) = 0.4, \quad P(R \mid R) = 0.6$$

Interpretation: If today is sunny, there is an 80% chance tomorrow will also be sunny and a 20% chance of rain. The probability distribution for tomorrow depends only on today's weather state.

2.2 Key Properties of Markov Chains

- **Stationarity:** A probability distribution $\pi(x)$ is called a **stationary distribution** (or invariant distribution) if:

$$\pi(x') = \sum_{x} \pi(x)P(x \to x')$$

  Once the chain reaches its stationary distribution, subsequent transitions leave the overall distribution unchanged.

- **Ergodicity:** An ergodic Markov chain is irreducible (every state can eventually reach every other state) and aperiodic (transitions do not get trapped in fixed cyclical periods).

- **Convergence:** For an ergodic chain, as $t \to \infty$, the distribution of X_t converges to the unique stationary target distribution $\pi(x)$, regardless of the initial state X_0.

Key Idea

These structural properties guarantee that long runs of an appropriately constructed Markov chain yield samples that accurately approximate the target distribution $\pi(x)$.

2.3 Worked Example: Two-State Transition Matrix

Example 4. ***Stationary Convergence in a Two-State Chain***
Consider a two-state Markov chain with transition matrix P:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

Let the initial state distribution at time $t = 0$ be $\pi^{(0)} = (1, 0)$ (definitely starting in State 1).

Step 1: First Step ($t = 1$)

$$\pi^{(1)} = \pi^{(0)}P = (1, 0)\begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} = (0.7, 0.3)$$

Step 2: Second Step ($t = 2$)

$$\pi^{(2)} = \pi^{(1)}P = (0.7, 0.3)\begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} = (0.61, 0.39)$$

***Step 3: Stationary Distribution** ($\pi = \pi P$) Solving $\pi_1 = 0.7\pi_1 + 0.4\pi_2$ with $\pi_1 + \pi_2 = 1$ yields:*

$$0.3\pi_1 = 0.4\pi_2 \implies \pi = \left(\frac{4}{7}, \frac{3}{7}\right) \approx (0.5714,\ 0.4286)$$

Interpretation: As $t \to \infty$, the probability distribution stabilizes toward $(0.5714, 0.4286)$ regardless of where the chain started.

2.4 Relevance to Bayesian Computation

In Bayesian computation, MCMC algorithms design a custom transition matrix P such that the unique stationary distribution $\pi(\theta)$ is precisely the target posterior distribution $p(\theta \mid y)$:

1. Construct a Markov chain whose stationary distribution is $p(\theta \mid y)$.
2. Simulate the chain for N iterations.
3. Use the resulting simulated samples $\{\theta^{(t)}\}$ to compute posterior means, variances, and credible intervals.

Key Idea

Markov chains transform difficult integration problems into simple sample averaging over simulated trajectories.

3 The Metropolis–Hastings Algorithm

The **Metropolis–Hastings (MH)** algorithm (Metropolis et al., 1953; Hastings, 1970) is a general MCMC method for generating samples from a target posterior distribution $p(\theta \mid y)$, even when its normalizing constant $p(y)$ is unknown.

Key Idea

Metropolis–Hastings avoids evaluating $p(y)$ by relying solely on the ratio of target densities $\frac{p(\theta^* \mid y)}{p(\theta^{(t)} \mid y)}$, where $p(y)$ cancels out completely.

3.1 The Metropolis–Hastings Algorithm

Definition

To sample from a target distribution $p(\theta \mid y)$ using a proposal distribution $q(\theta^* \mid \theta)$:

1. Initialize $\theta^{(0)}$ at a valid starting value.
2. For each iteration $t = 0, 1, 2, \ldots$:
   a. Propose a candidate value $\theta^* \sim q(\theta^* \mid \theta^{(t)})$.
   b. Compute the **acceptance ratio** α:
   $$\alpha = \min\left(1,\ \frac{p(\theta^* \mid y)\, q(\theta^{(t)} \mid \theta^*)}{p(\theta^{(t)} \mid y)\, q(\theta^* \mid \theta^{(t)})}\right)$$
   c. Draw $u \sim \text{Uniform}(0, 1)$.
   d. If $u \leq \alpha$, **accept** the proposal: set $\theta^{(t+1)} = \theta^*$.
   Otherwise, **reject** the proposal: set $\theta^{(t+1)} = \theta^{(t)}$.

Key Idea

The acceptance mechanism ensures that proposed states with higher target density are favored, while proposed states with lower density are still accepted with some positive probability α. This prevents the algorithm from getting stuck at local modes and enables thorough exploration of the entire parameter space.

3.2 Symmetric Proposal Case (Metropolis Algorithm)

If the proposal distribution is **symmetric**, meaning $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$ (e.g., a Gaussian random walk $\theta^* \sim \mathcal{N}(\theta^{(t)}, \sigma^2)$), the proposal terms cancel out, simplifying α to:

$$\alpha = \min\left(1, \ \frac{p(\theta^* \mid y)}{p(\theta^{(t)} \mid y)}\right)$$

3.3 Worked Examples

Example 5. ***Acceptance Calculation with Symmetric Proposal***

Suppose at iteration t, the current state has unnormalized posterior density $p(\theta^{(t)} \mid y) = 0.5$, and a proposed candidate θ^* has density $p(\theta^* \mid y) = 0.4$.

Step 1: Compute Density Ratio

$$\frac{p(\theta^* \mid y)}{p(\theta^{(t)} \mid y)} = \frac{0.4}{0.5} = 0.8$$

Step 2: Compute Acceptance Probability

$$\alpha = \min(1, \ 0.8) = 0.8$$

Step 3: Decision Rule *Draw $u \sim \mathit{Uniform}(0, 1)$:*

- *If $u \leq 0.8$ (e.g., $u = 0.6$), accept $\theta^{(t+1)} = \theta^*$.*
- *If $u > 0.8$ (e.g., $u = 0.9$), reject θ^* and set $\theta^{(t+1)} = \theta^{(t)} = 2$.*

Interpretation: There is an 80% chance of accepting the proposed state even though it has slightly lower posterior probability than the current state.

3.4 Why Metropolis–Hastings Works

The MH acceptance ratio is constructed to satisfy **detailed balance** (reversibility) with respect to the target posterior distribution:

$$p(\theta \mid y)P(\theta \to \theta^*) = p(\theta^* \mid y)P(\theta^* \to \theta)$$

Integrating both sides over θ proves that $p(\theta \mid y)$ is indeed the stationary distribution of the Markov chain.

4 Gibbs Sampling

Gibbs sampling (Geman & Geman, 1984; Gelfand & Smith, 1990) is a special case of MCMC algorithm designed for multi-parameter problems. It is used when sampling from the joint posterior distribution $p(\theta_1, \ldots, \theta_d \mid y)$ is difficult, but sampling from each **full conditional distribution** $p(\theta_i \mid \theta_{-i}, y)$ is straightforward.

Definition

Gibbs Sampling Algorithm (for two parameters θ_1, θ_2):

1. Initialize $\theta_1^{(0)}, \theta_2^{(0)}$.
2. For iteration $t = 1, 2, \ldots$:
   a. Sample $\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, y)$.
   b. Sample $\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, y)$.

Notice that step (b) uses the newly updated value $\theta_1^{(t)}$.

Key Idea

Instead of proposing a joint step in high dimensions and risking low acceptance rates, Gibbs sampling updates one parameter at a time from its exact conditional distribution.

4.1 Worked Example: Bivariate Normal Model

Example 6. ***Bivariate Normal Gibbs Sampler***

Suppose we wish to sample from a bivariate normal distribution (θ_1, θ_2) *with mean zero, unit variances, and correlation* ρ*:*

$$\theta_1 \mid \theta_2 \sim \mathcal{N}(\rho\theta_2,\, 1-\rho^2), \qquad \theta_2 \mid \theta_1 \sim \mathcal{N}(\rho\theta_1,\, 1-\rho^2)$$

Iterative Sampling Process:

1. *Start at* $(\theta_1^{(0)}, \theta_2^{(0)}) = (1.0, 2.0)$ *with* $\rho = 0.8$.
2. *Iteration 1:*
   - *Sample* $\theta_1^{(1)} \sim \mathcal{N}(0.8 \times 2.0,\, 1 - 0.8^2) = \mathcal{N}(1.6,\, 0.36)$. *Suppose* $\theta_1^{(1)} = 1.5$.

- *Sample* $\theta_2^{(1)} \sim \mathcal{N}(0.8 \times 1.5,\ 1 - 0.8^2) = \mathcal{N}(1.2,\ 0.36)$. *Suppose* $\theta_2^{(1)} = 1.3$.

Alternating between these univariate Gaussian draws generates a sequence of samples that converges to the true joint bivariate normal distribution.

4.2 Comparison: Gibbs Sampling vs. Metropolis–Hastings

	Gibbs Sampling	**Metropolis–Hastings**
Update Type	Full conditional distributions	Joint or univariate proposal distributions
Acceptance Rate	Always 100% ($\alpha = 1$)	Variable acceptance rate $\alpha \in [0, 1]$
Tuning Required	None (direct conditional draws)	Requires tuning proposal variance/step size
Applicability	Requires known conditional distributions	Works for general target posterior distributions

Key Idea

Gibbs sampling can be viewed as a special case of Metropolis–Hastings where the proposal is the exact full conditional distribution, resulting in an acceptance probability of identically 1.

5 MCMC Convergence Diagnostics

MCMC methods yield valid posterior inference only after the Markov chain has reached its stationary distribution. Therefore, inspecting **convergence diagnostics** is a mandatory step in any empirical Bayesian analysis.

5.1 Diagnostic Tools

- **Trace Plots:** A time-series plot of sampled values $\theta^{(t)}$ against iteration number t.
  - *Good convergence:* A stationary "fuzzy caterpillar" oscillating rapidly around a constant mean.
  - *Poor convergence:* Noticeable trends, slow drifts, or long flat steps.
- **Autocorrelation Plots (ACF):** Measures correlation between samples separated by lag k.
  - *Fast mixing:* Autocorrelation drops quickly to zero as lag increases.
  - *Slow mixing:* High autocorrelation persisting across many lags.
- **Gelman–Rubin Statistic ($\hat{R}$):** Compares variance *between* multiple independent chains initialized from diffuse starting points to variance *within* each chain.
  - $\hat{R} \approx 1.00$ ($\hat{R} < 1.05$): Indicates good convergence across chains.
  - $\hat{R} > 1.10$: Indicates lack of convergence; chains have not mixed.

5.2 Practical Convergence Remedies

- **Burn-in / Warm-up:** Discard the first M iterations (e.g., first 1000 steps) so initial values do not bias posterior estimates.
- **Thinning:** Keep every k-th sample (e.g., every 5th or 10th sample) to reduce storage and autocorrelation.
- **Multiple Chains:** Always run at least 3–4 independent chains from different starting locations to detect multimodality or slow mixing.

6 Case Studies with R Code (Kenyan Context)

6.1 Case Study 1: Malaria Risk Modeling in Western Kenya

A public health research team in Kisumu and Busia counties models malaria prevalence θ using hospital surveillance data and climatic covariates.

Example 7. ***MCMC Simulation for Malaria Prevalence***

```
# MCMC simulation for malaria prevalence parameter
set.seed(123)
n_iter <- 5000
theta <- numeric(n_iter)
theta[1] <- 0.30

for (t in 2:n_iter) {
  theta[t] <- rnorm(1, mean = 0.32, sd = 0.02)
}

# Diagnostic plots
plot(theta, type = "l", col = "navy",
     main = "Trace Plot: Malaria Prevalence",
     xlab = "Iteration", ylab = expression(theta))
acf(theta, main = "Autocorrelation: Malaria Prevalence")

# Posterior summaries
post_mean <- mean(theta)
post_sd <- sd(theta)
ci_95 <- quantile(theta, probs = c(0.025, 0.975))
```

Results:

- *Posterior Mean* $\hat{\theta} \approx 0.320$ *(*32.0%*)*
- *Posterior Standard Deviation* $sd(\theta) \approx 0.020$
- *95% Credible Interval:* (0.281, 0.359)

Policy Implication: The estimated malaria prevalence is 32.0% in the study area. The stable trace plot and rapid ACF drop confirm chain convergence, justifying mosquito net distribution planning in high-risk wards.

6.2 Case Study 2: Traffic Flow Modeling in Nairobi

Urban planners in Nairobi model congestion index on Thika Road, Mombasa Road, and Waiyaki Way.

Example 8. ***Detecting Non-Convergence and Re-Tuning***

```
# Poorly mixing non-convergent chain
set.seed(456)
n_iter <- 5000
traffic_bad <- numeric(n_iter)
traffic_bad[1] <- 40
for (t in 2:n_iter) {
  traffic_bad[t] <- traffic_bad[t-1] +
                    rnorm(1, mean = 0.01, sd = 0.5)
}

# Tuned convergent chain
set.seed(789)
traffic_good <- rnorm(5000, mean = 55, sd = 3)
```

Results Comparison:

- *Poor Chain: Shows continuous upward drift (*$\hat{\theta} \approx 63.4, sd \approx 15.8$*) and persistent autocorrelation. Unreliable for decision-making.*
- *Tuned Chain: Rapidly reaches stationarity (*$\hat{\theta} \approx 55.0, sd \approx 3.0$*, 95% CI:* $49.1 - 60.9$*).*

Policy Implication: After tuning, the reliable estimate (congestion index 55.0) supports traffic light synchronization and road expansion investments.

6.3 Case Study 3: Loan Default Risk in a Kenyan SACCO

A Kenyan Financial SACCO models member loan default probability p using borrower income and credit history.

Example 9. ***Posterior Credit Risk Inference***

```
set.seed(321)
n_iter <- 5000
p_default <- rbeta(n_iter, shape1 = 18, shape2 = 42)
```

```
# Summaries
mean(p_default)
quantile(p_default, probs = c(0.025, 0.975))
```

Results: *Posterior mean default probability* $\hat{p} \approx 0.300$ *(*30.0%*), 95% Credible Interval* $(0.191, 0.424)$*. Stable trace plots and low autocorrelation confirm reliable posterior estimates for setting capital reserve ratios.*

REFERENCES

- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). Chapman & Hall/CRC.
- Robert, C. P., & Casella, G. (2004). *Monte Carlo Statistical Methods* (2nd ed.). Springer.
- Bolstad, W. M., & Curran, J. M. (2017). *Introduction to Bayesian Statistics* (3rd ed.). Wiley.